Optimal-transport targets for macromolecular model fitting: non-vanishing gradients from a structure-factor computation

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Abstract

Real-space refinement targets are local: their gradient decays with the overlap between model and map, so a model placed even a few resolution widths away receives no usable signal. Optimal-transport (OT) discrepancies do not share this behaviour — the gradient of a Wasserstein-1 objective is bounded away from zero wherever mass is misplaced — but the formulations used to date represent both densities as point clouds in real space, which does not compose with refinement engines built on structure factors. We show that sliced W_1 against a sampled map *is* a structure-factor computation: the target's projected profile along any direction is a central slice of its transform, the model's projected profile has an exact one-dimensional transform which is a structure factor with the usual atomic form factor, and convolution, differentiation and integration all become reciprocal-space multipliers. Atomic gradients then follow by the route crystallography already uses, with atoms gridded at a common width and the residual B applied as a per-type multiplier. Against a case with a closed-form answer the gradient agrees with finite differences to 3×10^{-9}; the floor reaches 5×10^{-7} on cubic, non-cubic, monoclinic and triclinic grids alike; and the gridded route is $152\times$ faster than direct summation at 8×10^3 atoms. We argue that the resulting target is a *search* device to be followed by a conventional forward-model refinement, and we are explicit that the protocols needed to make it robust in production are outside the scope of this work.

1 Introduction

1.1 Why refinement targets go blind

A real-space refinement target scores a model by how its computed density agrees with an observed map, and every such score is built from products of the two densities evaluated where they overlap. When a model is misplaced by more than a resolution width that overlap falls off like the tail of a Gaussian, and with it the gradient. The practical consequence is familiar: a ligand modelled into the wrong lobe of density, or a side chain in the wrong rotamer, is not repaired by refinement, because the target has nothing to say about where the model should go. It has to be moved by something else — a human, a search, or a docking calculation — and refinement then polishes whatever it is handed.

This is not a defect of any particular target so much as a property of local comparison. A least-squares residual on the map, a correlation coefficient, and the sum of density at atomic centres all share it, and they degrade in two distinguishable ways (§4): the sum-of-density gradient keeps pointing roughly the right way but shrinks by nine orders of magnitude over ten Ångström, while the least-squares gradient keeps its magnitude and *reverses direction* once the two densities cease to overlap and the model's self-term dominates.

1.2 Optimal transport, for crystallographers

Optimal transport supplies a different way to compare two densities, and one that does not require them to overlap at all. Given two non-negative densities of equal total mass, the Kantorovich problem asks for the cheapest way to rearrange one into the other, where the cost of moving a unit of mass from x to y is some ground cost $c(x, y)$. Taking $c(x, y) = \|x - y\|$ gives the Wasserstein-1 distance W_1, sometimes called the earth-mover's distance: the answer is literally how far mass has to travel, and it is finite and informative however far apart the two densities are.

The reason this is not routine in crystallography is cost. Solving a transport problem between two three-dimensional maps is a large linear program, and the entropically regularised solvers that made OT tractable in machine learning introduce a regularisation parameter whose setting is itself a problem. The device that makes OT affordable is *slicing*. In one dimension the transport problem has a closed-form solution obtained by sorting: the optimal map is the monotone rearrangement, and

$$W_1(\mu,\nu) = \int \big|F_\mu(t) - F_\nu(t)\big|\,\mathrm{d}t \tag{1}$$

where F denotes a cumulative distribution. Sliced OT replaces the d-dimensional problem by an average over one-dimensional problems obtained by projecting both measures onto a set of directions,

$$\mathcal{SW}_1(\mu,\nu) = \frac{1}{L}\sum_{\ell=1}^{L} W_1\big(P_{u_\ell}\mu,\, P_{u_\ell}\nu\big), \tag{2}$$

each of which is a sort. The result is a genuine metric, it inherits the topology of W_1, and it costs $O(N \log N)$ per direction rather than the cost of a linear program.

Sliced OT is by now well established in imaging and increasingly in cryo-EM. AlignOT [cite Riahi et al., IEEE/ACM TCBB 20, 3842 (2023)] aligns two EM maps represented as point clouds by minimising a Wasserstein objective over rotations, reporting convergence from a wider range of starting angles than local alignment. EMPOT [cite Riahi et al., PRX Life 3, 023003 (2025)] extends this to partial overlap and rigid-body fitting of atomic models using an unbalanced Gromov–Wasserstein divergence. Related work uses transport for interpolation between maps [MorphOT], for manifold learning over conformational ensembles [Zelesko et al., ISBI 2020], and for three-dimensional reconstruction [CryoSWD, Zehni & Zhao, ICASSP 2023]. Most directly, a recent study [bioRxiv 2025.12.23.696001] reports that a sliced-Wasserstein image loss "significantly broadens the basin of attraction" relative to mean-squared error in cryo-EM optimisation. *That OT opens gradients is therefore established*, and it is not what this paper claims. What is missing is a formulation that a crystallographic refinement engine can actually execute.

1.3 Non-vanishing gradients: the good and the bad

The property that makes W_1 attractive follows from its dual. The optimal potential is 1-Lipschitz and, in one dimension, has the closed form

$$f'_\ell(t) = -\operatorname{sgn}\big(F_{P_{u_\ell}\mu}(t) - F_{P_{u_\ell}\nu}(t)\big), \tag{3}$$

a *sign*. Its magnitude is one wherever the two cumulative distributions differ at all, and it is defined everywhere rather than only where the densities overlap. Smoothing by the atomic form factor and back-projecting gives the atomic gradient (§2), whose magnitude is therefore bounded below as long as any mass is misplaced, at any separation.

That is exactly the wanted behaviour for a search, and exactly the wrong behaviour for a refinement. Because the gradient is a smoothed sign, it does not diminish as the model approaches the answer, and it therefore carries no natural step length: a descent scheme driven by it overshoots and limit-cycles rather than converging. Two consequences follow and we adopt both. First, the step length must come from somewhere else; the barycentric projection of the same transport plan — the monotone rearrangement itself — supplies a displacement rather than a force, and it *does* vanish at the solution (§5). Second, and more importantly, the transport term should not be the final target at all. Left unrestrained it will happily distort a model to fit density: we measure a configuration with 1.25 Å distance errors scoring $8.5\times$ better on the transport term than the chemically exact answer. The intended role is to deliver a model into the correct basin, after which a conventional forward-model refinement — with its proper treatment of scaling, bulk solvent, and stereochemistry — takes over.

1.4 Why not just search harder

The obvious alternative to a better target is a better search: exhaustive rotation– translation scans, docking, or simulated annealing. These work, and for rigid-body placement of a whole molecule they are standard. The difficulty is atomic-level fitting, where the search space is not six-dimensional but $3N$-dimensional, and where the interesting failures — a wrong rotamer, a peptide flip, a ligand bound in a shifted register — are not reachable by rigid-body moves. Conformer-ensemble methods address this by enumerating plausible geometries in advance, but the number of conformers grows quickly and each must still be placed and scored. What is wanted is a target whose gradient reaches, so that a small number of starting points suffices, and which is cheap enough to evaluate that an ensemble of them can be scored at all. The formulations cited above are point-cloud based and rigid-body; they do not provide per-atom analytic gradients, and they scale poorly in the atom count where a protein fragment or a cofactor is concerned.

1.5 What this paper does

We show that sliced W_1 against a sampled map is a structure-factor computation, and that its atomic gradients can therefore be evaluated by the FFT route crystallography has used since Agarwal [cite Agarwal 1978] and Ten Eyck [cite Ten Eyck 1973/1977]: grid the atoms at a common width, apply the residual B as a per-type reciprocal-space multiplier, transform, and gather at the atomic positions. The contribution is the formulation and its cost, not an application. We demonstrate that naive, hand-set targets are already enough to recover badly placed models in two synthetic cases, and we are explicit that the protocols required to make this robust on experimental data — masking and segmentation decisions, per-atom B treatment, handling of negative density, symmetry mates and the surrounding structure — are identified but not solved here.

2 Formulation

[Slice theorem; exact 1-D structure factor for the model; operators as multipliers; the phase-origin and branch conventions. Text exists in the earlier draft — port it.]

3 Atomic gradients by the Agarwal/Ten Eyck route

[The B-factor split; forward scatter and backward gather; per-type grouping; the observation that the gather kernel is the form factor itself rather than its derivative, because the $2\pi iq$ from the phase cancels against the $1/2\pi iq$ of the spectral antiderivative.]

Table 1: Cost of value and gradient, $L = 24$, CPU, single precision-independent. K grows with the model because the projection window must contain it, so direct summation scales worse than linearly in N.

N	K	direct (ms)	gridded (ms)	speedup
23	95	2.6	3.9	0.7×
200	119	22.0	5.6	4.0×
2000	257	4594	64.7	71×
8000	467	26079	171.4	**152**×

4 Reach, and how conventional targets fail

Capped leucine fragment (13 heavy atoms, N-acetyl-leucine-N'-methylamide), 2.5 Å, $\sigma = 1.062$ Å. "cos" is the cosine between $-\nabla E$ and the true displacement.

Radius of convergence, random rotation (0–30°) plus translation of magnitude d, 10 trials, identical restraint projection and identical normalised step for every target, no annealing, success at RMSD < 0.5 Å:

t (Å)	t/σ	OT		least squares		sum of density	
		$\|\|g\|\|/w$	cos	$\|\|g\|\|$	cos	$\|\|g\|\|$	cos
2	1.9	0.5009	0.996	1.6e-5	0.988	1.6e-5	0.992
4	3.8	0.5009	0.996	1.6e-5	0.976	2.8e-6	0.955
6	5.7	0.5009	0.996	1.4e-5	**0.431**	5.6e-8	0.914
8	7.5	0.5009	0.996	1.4e-5	**−0.280**	1.0e-10	0.895
12	11.3	**0.5009**	**0.996**	1.4e-5	0.229	1.6e-14	−0.962

d (Å)	d/σ	OT	least squares	sum of density
1–3	≤2.8	100%	100%	100%
4	3.8	**100%**	100%	90%
6	5.7	**100%**	90%	80%
8	7.5	**100%**	10%	0%

Normalising the step for every target deliberately favours the conventional ones, since it neutralises their magnitude decay and tests only direction; the advantage reported here is therefore a lower bound.

5 Step length from the barycentric projection

[M^{-1} backprojection factor, one-step recovery of a rigid translation, and the measured comparison: final RMSD 0.010 Å with δ-derived steps against 0.133 Å with a fixed cap, oscillation 0.002 against 0.057 Å.]

6 Validation

Table 2: End-to-end validation across grid geometries, leucine fragment, 2.5 Å, $L = 32$. The floor is a diagnostic of the *map*: it detects inadequate padding, and requires a clearance of $\gtrsim 7\sigma$ between the model and the box or cell boundary.

grid	floor	anchor	FD median	reach @ 12 Å	1-step
cubic 48^3	4.79e-7	+0.000%	1.3e-9	+0.000%	5.2e-3 Å
non-cubic, anisotropic step	4.76e-7	+0.000%	7.7e-10	+0.000%	5.2e-3 Å
monoclinic $60 \times 66 \times 54$	4.73e-7	+0.000%	1.7e-9	+0.000%	5.0e-3 Å
triclinic $96 \times 100 \times 104$	4.80e-7	+0.000%	2.7e-9	+0.000%	5.0e-3 Å

The formulation is exact on a general grid because the target slices are computed as a structure factor over voxels rather than by interpolating a central slice of the map's transform; the only place grid geometry enters is each voxel's Cartesian position. Bandwidth is set by the shortest reciprocal vector of the sampling lattice, which on a skewed cell can be twice what the step lengths suggest.

7 Demonstrations

[NOT YET RUN. Both use synthetic data, equal B factors, a naive hand-set stage schedule, the alternating-projection protocol (P_{data} then P_{restr}, over- relaxed), and a final map-correlation refinement. In both cases the same protocol started from a map-correlation target is the control.]

7.1 Pentapeptide with wrong rotamers and backrub main-chain motion

[Design: build a pentapeptide, perturb two side chains to incorrect rotamers and apply a backrub motion to the intervening main chain; refine; report RMSD by residue and separately for main chain and side chain.]

7.2 NAD$^+$/NADH from a conformer ensemble

[Design: generate a conformer ensemble, place each into the map of the correct conformer, score by transport, refine the best; report rank of the correct conformer and final RMSD.]

8 Scope and what is deliberately left out

[Search target not refinement target; balanced formulation and what breaks for a fragment in a larger assembly; negative density and the segmentation decision (measured: contouring truncates the second moment by up to 25% and degrades the fit by $6\times$); per-atom B; symmetry mates and environment. Electron scattering factors are also Gaussian sums, so the same route serves cryo-EM.]